

MCHE_Designer User's Manual

V1.1.0.0

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MCHE Designer User Manual

1 Introduction

This paper is the user manual for MCHDesigner V1.1.0.0. The software is free of installation and can be used directly after downloading and decompressing it.

MCHE Designer is a software dedicated to the design of micro-channel heat exchangers in the SanHua Micro-Channel Software Package(SanHua MCSP). Users can double-click the "MCHDesigner.exe" executable program in "./SanHua MCSP/HX Sim/".

2 MCHDesigner Features

The "MCHDesigner.exe" is used for the simulation and design of MCHE. Its main functions include:

- evaporator/condenser simulation with uniform and non-uniform air mass flow and refrigerant flow;
- Interlaced MCHE simulation;
- MCHE simulation with parallel refrigerant flow;
- Simulation of non-uniform port tube heat exchanger.

Through "Tool->Register" in the menu bar, user can associate ".sh" files with software. Before the first time successfully opening the software via double clicking ".sh" file, user may need to right-click ".sh" file and choose to open it with the current software.

2.1 Main UI

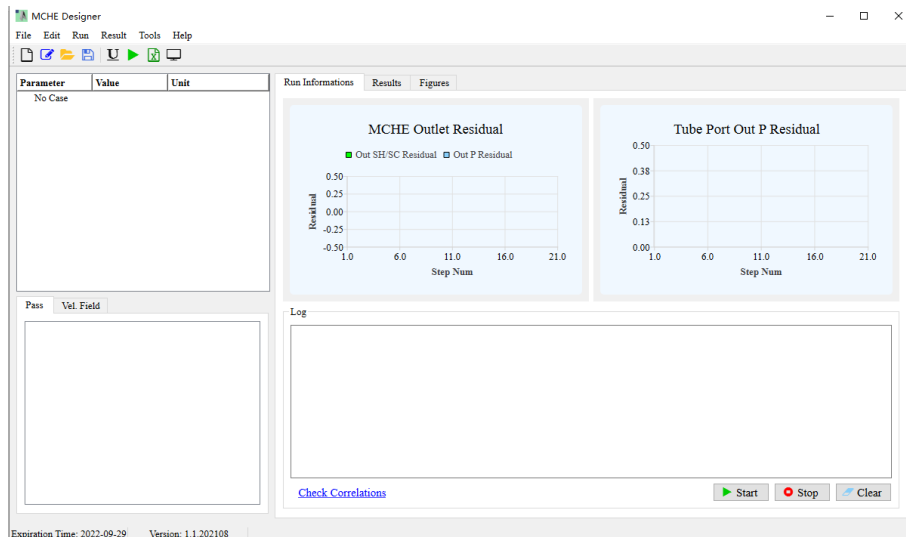


Figure 2.1 Main UI

Figure 2.1 shows the main UI of the software. The left side of UI shows the input data and refrigerant flow pass. When the input is the non-uniform air flow field, user can select the "Vel. Field" page to view the flow velocity cloud. The status bar in the lower left corner shows the expiration time and the software version. The right side displays the calculation process.

The iterative process will be dynamically displayed during the simulation.

2.2 Toolbar

Figure 2.2 shows the software toolbar. From left to right are “New”, “Edit”, “Open”, “Save”, “Unit Setting”, “Run”, “Save to Excel” and “Window Size Reset”.

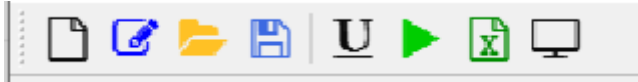


Figure 2.2 Toolbar

The “New” and “Edit” will be described in detail in section 3. The “Window Size Reset” is used to restore the default UI size.

The “Unit Setting” in the toolbar is to uniformly modify the parameter units. The UI is shown in Figure 2.3.

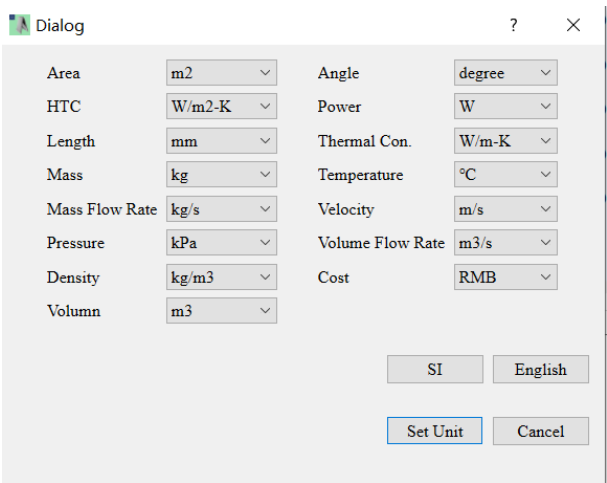


Figure 2.3 Unit Settings

2.3 Result

The simulation result can be shown by selecting the "Result" page on the right side of the main UI, as shown in Figure 2.4. The results show in standard units and imperial units. The user can adjust the result unit by clicking the "Unit" button.

To export the result, click the "Result" in the menu bar, as shown in Figure 2.5. The current output files include excel files and PDF file. Among them, “toExcel (CD Template)” indicates that the excel file format is based on the commercial software template, which is not suitable for non-uniform geometry cases.

Run Informations Results Figures						
Performance						
Parameter	Value	Unit	SI Value	SI Unit	Eng. Value	Eng. Unit
Heat Loads						
Total Heat Load	15964.392	W	15964.392	W	54472.767	Btu/hr
Sensible Heat Load	12188.312	W	12188.312	W	41588.246	Btu/hr
Latent Heat Load	3776.080	W	3776.080	W	12884.521	Btu/hr
Sensible Heat Ratio	0.763		0.763		0.763	
Ref. Liquid Heat Load	0.000	W	0.000	W	0.000	Btu/hr
Ref. Two Phase Heat Load	15448.604	W	15448.604	W	52712.825	Btu/hr
Ref. Vapor Heat Load	515.788	W	515.788	W	1759.942	Btu/hr
Charge						
Total Refrigerant Charge	0.049	kg	0.049	kg	0.107	lbm
In-tube Ref. Liquid Charge	0.000	kg	0.000	kg	0.000	lbm
In-tube Ref. Two Phase Charge	0.047	kg	0.047	kg	0.104	lbm
In-tube Ref. Vapor Charge	0.002	kg	0.002	kg	0.003	lbm
Header Charge	0.000	kg	0.000	kg	0.000	lbm
Flow Rates						
Total Refrigerant Flow Rate	355.045	kg/hr	0.099	kg/s	13.046	lbm/min
Air Mass Flow Rate	3435.465	kg/hr	0.954	kg/s	126.232	lbm/min
Air Flow Rate	0.825	m ³ /s	0.825	m ³ /s	1748.252	ft ³ /min
Standard Air Flow Rate	0.820	m ³ /s	0.820	m ³ /s	1738.063	ft ³ /min
Air Velocity	1.655	m/s	1.655	m/s	325.813	ft/min
Pressure Drops						
Total Air Side Pressure Drop	26.627	Pa	26.627	Pa	0.004	psi
Total Air Side Pressure Drop on DB	0.000	Pa	0.000	Pa	0.000	psi
Refrigerant Side Pressure Drop	13925.122	Pa	13925.122	Pa	2.020	psi
Refrigerant Saturation Temperature Drop	0.543	K	0.543	K	0.978	°F
Inlet Header Ref Side Pressure Drop	0.000	Pa	0.000	Pa	0.000	psi
Outlet Header Ref Side Pressure Drop	0.000	Pa	0.000	Pa	0.000	psi

Figure 2.4 Result

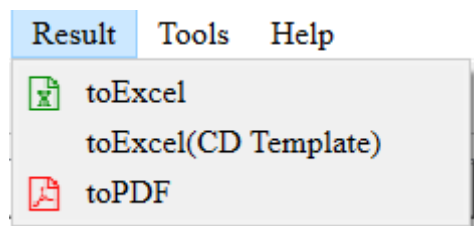


Figure 2.5 Result Export

2.4 Unit Preference

The user can set the unit preference via clicking “File->Preference” on menu bar. The UI is shown in Figure 2.6.

This function enables the user to set the default unit of the parameter when creating a new case. At the same time, after the calculation is completed, without setting the output result unit, it is displayed based to the preference setting.

It should be noted that the unit setting on the toolbar has a higher priority and will override the preference setting.

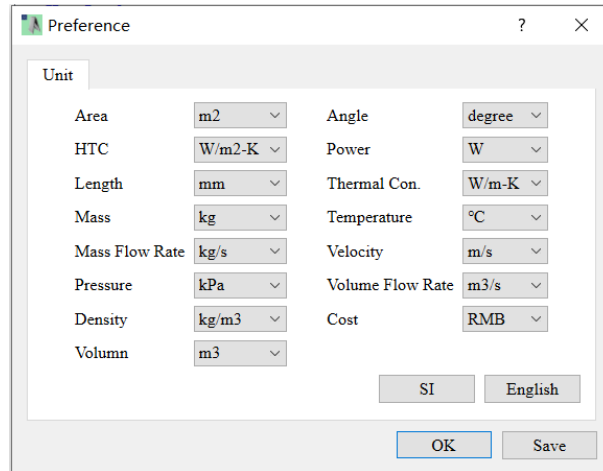


Figure 2.6 Unit Preference Settings

2.5 Interlaced MCHE

As shown in Figure 2.7, the current software supports Interlaced MCHE simulation. The user can check the "Interlaced MCHE" option in the input UI to enable this function. At this condition, the geometry and boundary condition parameter settings are consistent with the conventional one. During the simulation process, the model will automatically shield half of the tubes, which are treated as fins.

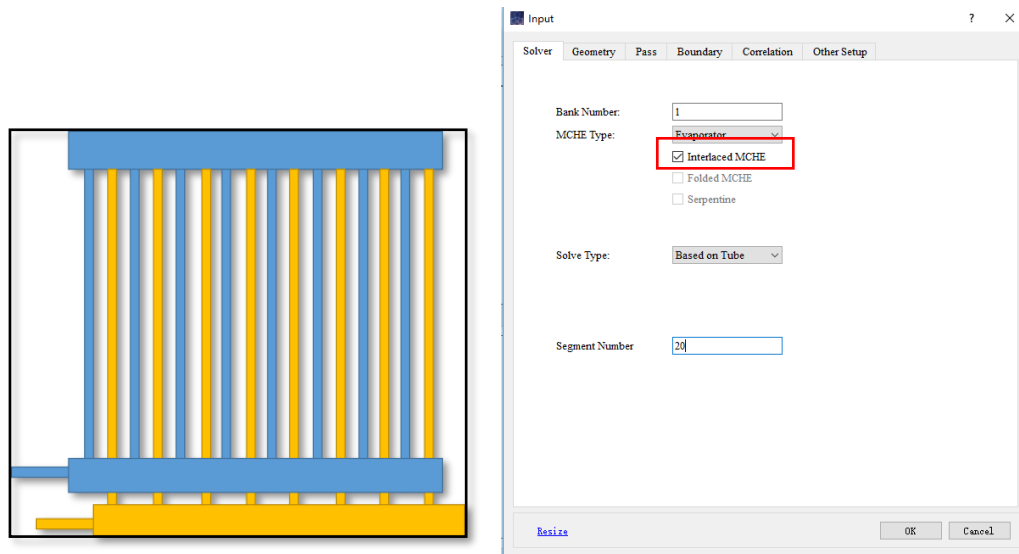


Figure 2.7 Interlaced MCHE

2.6 Non-Uniform Geometry Setting

As shown in Figure 2.8, the current software supports non-uniform geometry data input for tubes and fins in multi-banks MCHE. Users can enable this function by unchecking "Uniform Tube Gem." and "Uniform Fin Gem.".

There are three types for non-uniform geometry. After completing the data input, user need to select the corresponding type in "Other Setup" (as shown in Figure 2.9), including:

1. The front and rear banks have the same windward area;
2. Folded MCHE, the length of the tube is inconsistent after folded;
3. For the parallel refrigerant flow, the rear bank area is smaller than the front one. The relative position need to be defined (UDL). The location definition "along Tube Length"/"along Tube Height" is shown in Figure 2.9.

When the windward area of front and rear bank are inconsistent (the second and third type), it is currently only applicable to double-banks MCHE, and it is necessary to ensure that the windward area of the first bank is larger than the rear one.

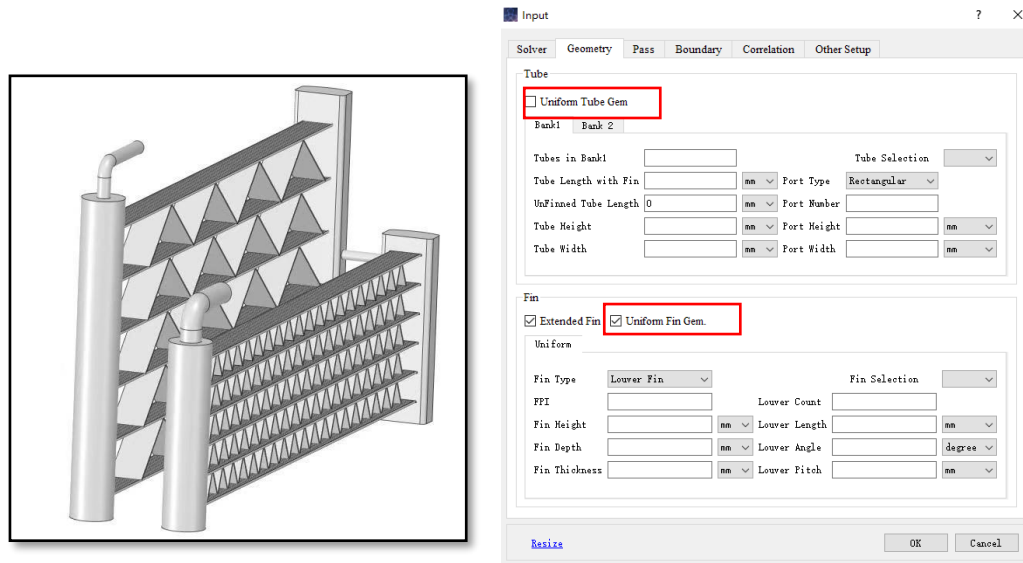


Figure 2.8 Geometry Input

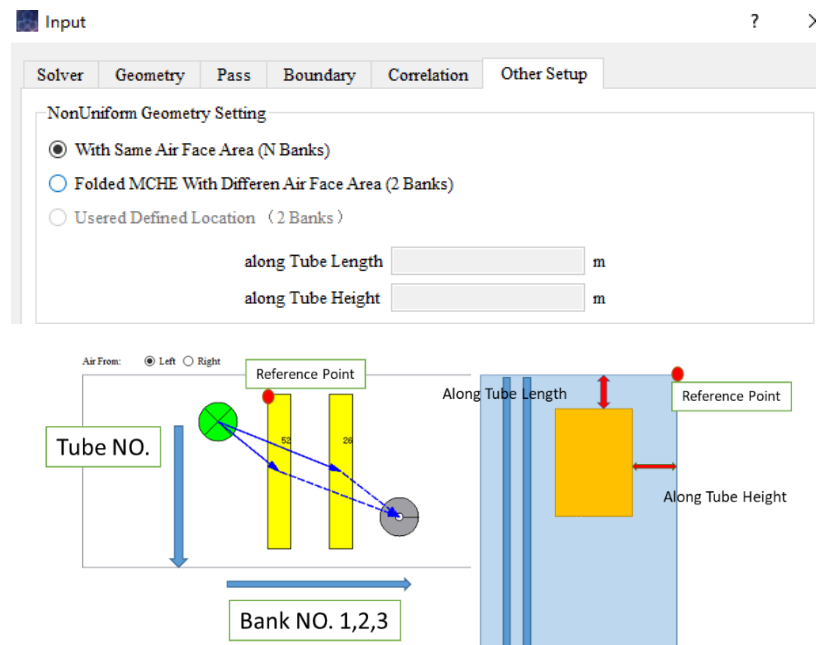


Figure 2.9 Non-Uniform Geometry Type

2.7 Parallel Refrigerant Flow

The current software supports the simulation of parallel refrigerant flow. The definition

of parallel refrigerant flow is shown in Figure 2.9. Users need to set the corresponding flow type in the “Refrigerant” page (Figure 2.10):

- 1. The outlet pressure drop and SH are the same;
- 2. The outlet pressure is the same, but the SH is inconsistent.

SolverGeometryPassBoundaryCorrelationOther Setup

Air SideRefrigerant Side

Refrigerant Side

RefrigerantR134aFastSolveTypeInlet P+X+M

TXV/EXV Inlet

PressurekPaTemperature°C

Heat Exchange Inlet

Sat. T°CSH(+)°CPressurekPaTemperature°CMass Flowkg/sQuality

Heat Exchange Outlet

PressurekPaSH(+) / SC(-)°C

☐ Ref. in Single Flow

☒ Ref. in Parallel Flow with Same Out SH and Out P

☐ Ref. in Parallel Flow with Same Out P

Figure 2.10 Refrigerant Flow Type

2.8 Refrigerant Mal-distribution

After completing the data input, the user can customize the refrigerant inlet mass flow among tubes through “Edit->Ref. Distribution”, as shown in Figure 2.11, where the flow is based on percentage, and the inlet quality of each tube is not affected.

Ref. Distribution

☐ Uniform Distribution

☒ MalDistribution

Inlet 1Inlet 2

	Tube ID	Ref. M(%)
Tube_1	1	0
Tube_2	2	0
Tube_3	3	0
Tube_4	4	0
Tube_5	5	0
Tube_6	6	0
Tube_7	7	0

OK

Cancel

Figure 2.11 Refrigerant Mal-Distribution

2.9 Batch Run

After completing the benchmark case, user can open the batch run simulation setting UI through "Run->BatchRun" in menu bar, as shown in Figure 2.12. The batch run parameter

settings are consistent with commercial software. In addition, users can click "Load From Excel File" to import the Excel setting file (see “./SanHua MCSP/ Template_BatchRun-2G25-JC523.xlsx” for template), and the parameter setting is more free.

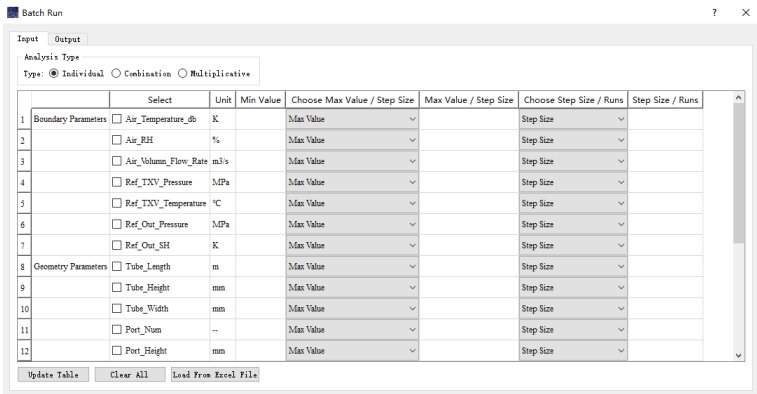


Figure 2.12 Batch Run

2.10 Manifold Model

The manifold model is supported for the simulation of inlet and outlet header pressure drop and the weight and cost of MCHE.

The geometry input is set after the flow pass. As shown in Figure 2.13, double-click the blue arrow to open the parameter input UI. When finishing the input, the arrow turns red. The user can click the arrow to view the input parameters in the upper right window, as shown in Figure 2.14.

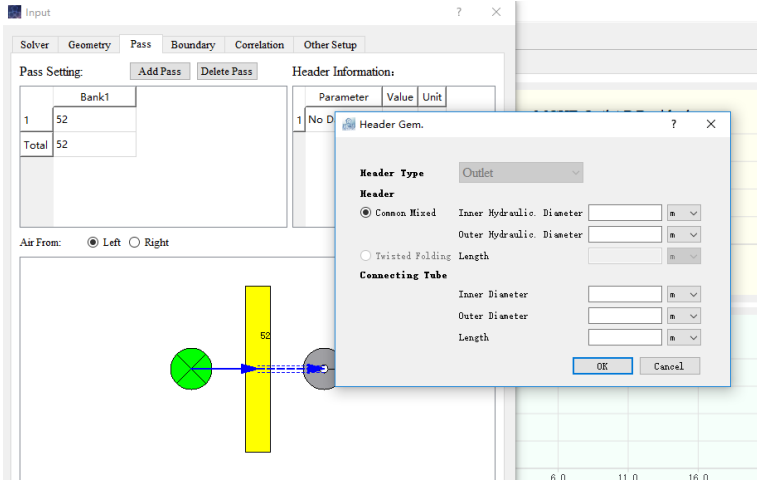


Figure 2.13 Header Geometry Input

Pass Setting:

	Bank1
1	52
Total	52

Header Information:

Parameter	Value	Unit
1 Header Type	Outlet	
2 Header I.D.	27	mm
3 Header O.D.	32	mm
4 Con. Tube I.D.	10	mm
5 Con. Tube O.D.	12	mm

Air From: ☒ Left ☐ Right

[Generate Pass](#)

Figure 2.14 Header Information

2.11 Weight/Cost Calculation

In “Input->Other Setup” UI, user can enter the density and price of the material for weight and cost calculation, as shown in Figure 2.15.

Materials Density

Tube	Al	2700	kg/m ³
Fin	Al	2700	kg/m ³
Header	Al	2700	kg/m ³
Connecting Tube	Copper	8940	kg/m ³

Cost

Tube	0	RMB/kg
Fin	0	RMB/kg
Header	0	RMB/kg
Connecting Tube	0	RMB/kg

Figure 2.15 Weight and Cost

2.12 Non-Uniform Port Distribution Simulation

The current version of MCHE Designer supports the simulation of non-uniform port distribution MCHE. After selecting the "Based on Port" solver, user can check the "Non-Uniform Port Distribution" option, as shown in Figure 2.16.

After enabled, user can click the "SetPortWidth" button on the geometry input UI to set the tube port width. Port1 in the table corresponds to the windward side, and for the folded heat exchanger Port1 is the outermost hole, seeing Figure 2.17 for details.

User can check "Set Nose/Web" and enter the wall thickness between channels in the table.

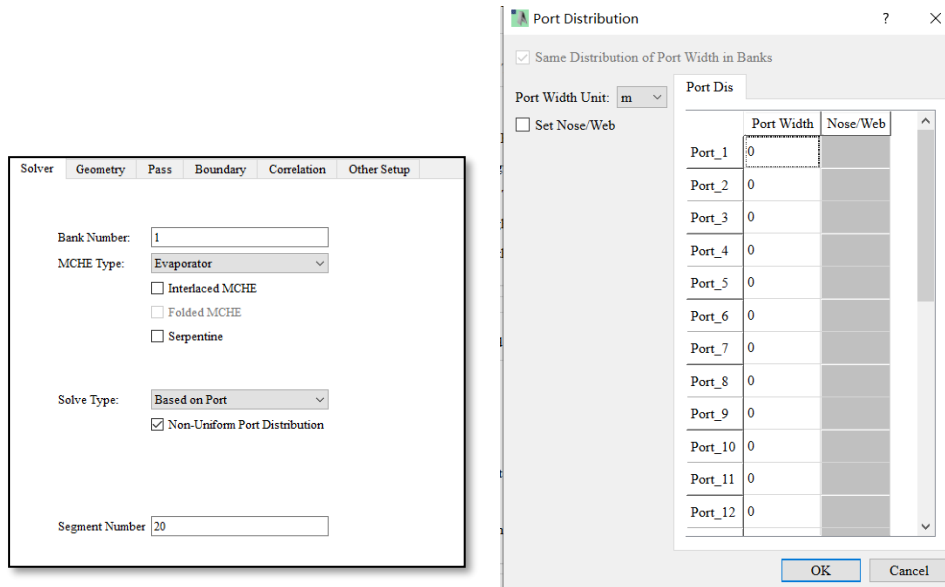


Figure 2.16 Non-Uniform Port Distribution

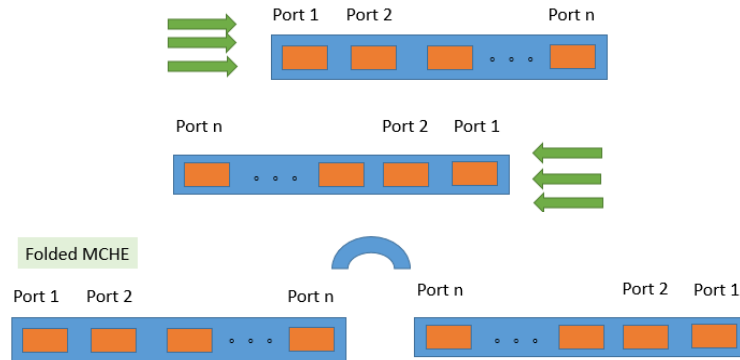


Figure 2.17 Port No.

2.13 Load CD(.chx) file

MCHE Designer supports to load the CD setting files (“.chx”). The user can click “File->Load CD(.chx) file” in the menu bar to load it, as shown in Figure 2.18. It should be noted that the CD file cannot be locked.

If there exists problems in the loading process, it will be written into the “./WarnInfo.txt”. After loading, do not forget to check if there exist any problems in the settings.

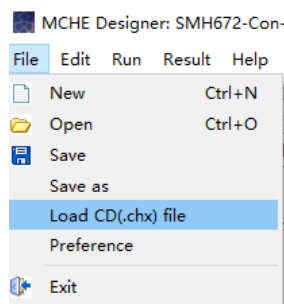


Figure 2.18 Load CD File

2.14 Add Refrigerant

User can add own refrigerants on the basis of the software provided. The refrigerant file is MIX and FLD format (NIST REFPROP file format). Put the file into the ".SanHua MCSP/User/UD_Ref " folder. In the setting UI, the added refrigerant can be found with the prefix "UD_" under the refrigerant drop-down menu, as shown in Figure 2.19.

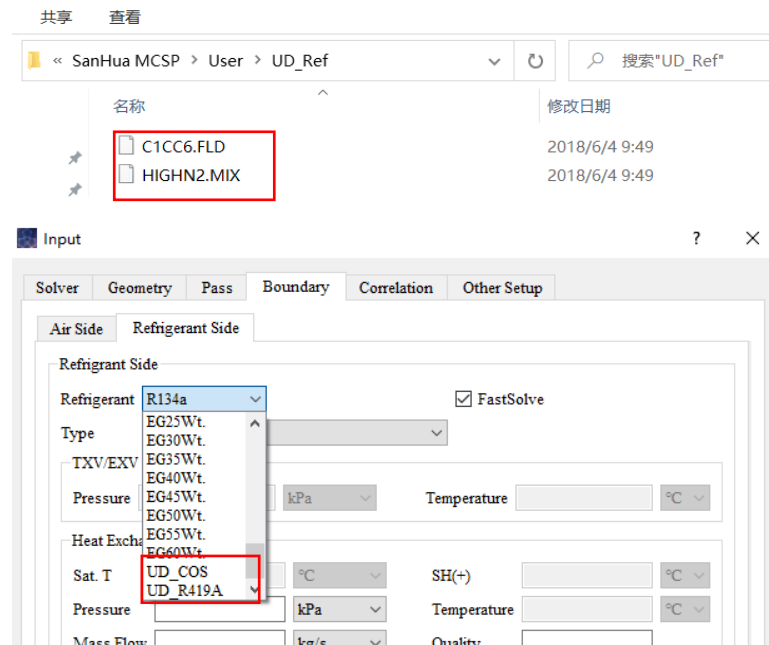


Figure 2.19 Custom Refrigerant

2.15 Port Blockage

When the "Based on Port" solver is turned on, the software can simulate port blockage. As shown in Figure 2.20, on the "Blockage" page, first enter the number of blocked tubes. And then enter the bank number, the tube number, and the port number (based on the windward direction) .

Blockage Tube Number:

	Bank No.	Tube No.	Port No.
1	1	1	1,2,3,4,5,6
2	1	2	1,2,3,4,5,6
3	1	3	1,2,3,4,5,6
4	1	4	1,2,3,4,5,6
5	1	5	1,2,3,4,5,6
6	1	6	1,2,3,4,5,6
7	1	7	1,2,3,4,5,6
8	1	8	1,2,3,4,5,6
9	1	9	1,2,3,4,5,6
10	1	10	1,2,3,4,5,6
11	1	11	1,2,3,4,5,6,7
12	1	12	1,2,3,4,5,6,7
13	1	13	1,2,3,4,5,6,7

Tips: The Port No. can be Split by "," such as "6,8,10".

Figure 2.20 Blockage

3 MCHE Designer Wizard

User can create a new project by clicking “File->New” on menu bare or "New" on toolbar. To modify an existing project, user can open the project through “File->Open” on menu bare or "Open" on toolbar and then edit it through “File->Edit” or "Edit". Figure 3.1 shows the parameter input UI. “Resize” in the lower left corner is to restore the initial size of the window.

Input

?

×

Solver

Geometry

Pass

Boundary

Correlation

Other Setup

Bank Number:

MCHE Type:

Evaporator

▼

☐ Interlaced MCHE

☐ Folded MCHE

☐ Serpentine

Solve Type:

Based on Tube

▼

☐ Non-Uniform Port Distribution

Segment Number

Resize

OK

Cancel

Figure 3.1 Input UI

3.1 Solver Page

In the “Solver” page, as shown in Figure 3.1, the following parameters are included:

1. Bank Number;
2. MCHE type: Evaporator or Condense;
3. Solver type;
4. Segments Number.

In the MCHE type, following choices are available:

- 1) Interlaced MCHE which is used for the simulation of Interlaced heat exchanger;
- 2) Folded MCHE which is used for multi-bank folded MCHE;
- 3) Serpentine which is under-development and not convenient for the ref. flow pass settings.

Types of solvers include: “Based on Tube” and “Based on Port”. The main difference is that the latter takes into account the non-uniform distribution of refrigerant among tube ports. Under “Based on Port”, the simulation of the non-uniform port distribution is available. Details can be found in section 2.12.

3.2 Geometry Page

In “Geometry” page, tube and fin geometry are input as shown in Figure 3.2. For multi-bank MCHE, different geometry parameters can be given among banks. The unit can be changed freely. The tube and fin in the database can be quickly selected through “Tube Selection” and “Fin Selection”. “UnFinned Tube Length” is the length of the tube without fins, including the part inserted into the header.

The screenshot shows the 'Input' dialog box with the 'Geometry' tab selected. The 'Tube' section has the following parameters:

- ☒ Uniform Tube Gem
- Uniform
 - Tubes per Bank: 52
 - Tube Length with Fin: 0.998 m
 - UnFinned Tube Length: 0 m
 - Tube Height: 1.3 mm
 - Tube Width: 25.4 mm
 - Tube Selection: [Dropdown]
 - Port Type: Rectangular
 - Port Number: 26
 - Port Height: 0.74 mm
 - Port Width: 0.635 mm

The 'Fin' section has the following parameters:

- ☒ Extended Fin ☒ Uniform Fin Gem
- Uniform
 - Fin Type: Louver Fin
 - FPI: 16
 - Fin Height: 8.1 mm
 - Fin Depth: 25.4 mm
 - Fin Thickness: 0.08 mm
 - Fin Selection: [Dropdown]
 - Louver Count: 12
 - Louver Length: 7.2 mm
 - Louver Angle: 45 degree
 - Louver Pitch: 1.85 mm

Buttons at the bottom: [Resize](#), OK, Cancel.

Figure 3.2 Geometry Input

3.3 Pass Page

In “Pass” page, users can specify the refrigerant flow in the MCHE, as shown in Figure

3.3. First, enter the tube number of each pass in each bank in the table. User can add or delete a pass by clicking "Add Pass" and "Delete Pass".

Then click the "Generate Pass" button to generate the pass matrix. The green circular icon is the inlet. The gray circular icon is the outlet. The yellow rectangle represents the flow pass and the number is the tube number in the pass.

The yellow icon cannot be moved, and the inlet and outlet icon can be moved by key "Shift" + left mouse button.

The blue arrows represent the refrigerant flow direction among the pass. Users specify the flow direction by clicking the upper icon with left mouse button, moving the mouse to the downward icon and then releasing the mouse. If users press key "D" and then move the mouse, the connecting line appears as a dotted line.

The solid line represents the refrigerant flowing from the outside to the inside; the dotted line represents the refrigerant flowing from the inside to the outside.

Right-click on the arrows shows the choices to delete the arrow.

"Left" and "Right" is the air flow direction.

After the pass settings, the header geometry can be input, seeing section 2.10 for details.

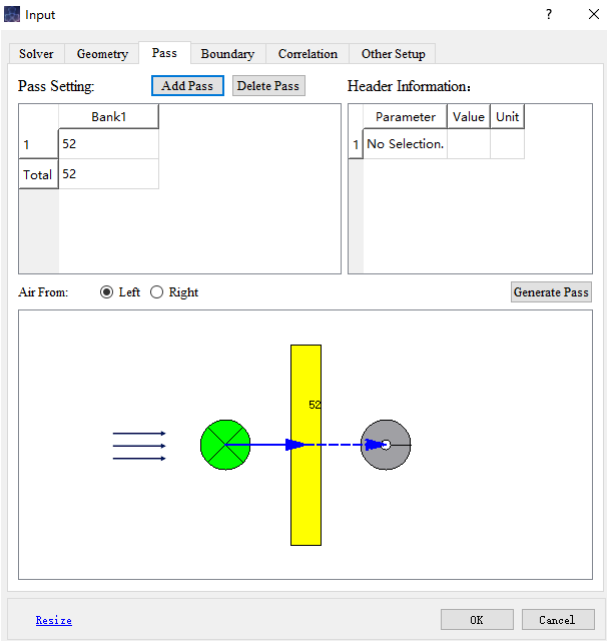


Figure 3.3 Flow Pass Settings

3.4 Boundary Page

Figure 3.4 shows the boundary condition settings, including the air side (a) and the refrigerant side (b).

The air side parameters mainly include: pressure, dry bulb temperature, wet bulb temperature or relative humidity, volume flow or velocity.

Select "NonUniform Velocity" to enable non-uniform velocity input, as shown in Figure 3.5. The velocity field can be imported from an Excel or Text file. For the file format, see the

"NonUniformVel_Temp.xlsx" and "NonUniformVel_Temp.txt" templates at folder “./SanHua MCSP/Template File”.

Input

Solver Geometry Pass Boundary Correlation

Air Side Refrigerant Side

Air Side

Pressure 101325 Pa

Dry Bulb Temperature 299.85 K

RH or Wet Temperature

☒ RH 50.36 %

☐ Wet Bulb Temperature K

Volum Flow Rate or Velocity

☒ Volum Air Flow Rate 0.825083 m³/s

☐ Air Velocity m/s

☐ NonUniform Velocity Set NonUniform

OK Cancel

(a) Air side

Air Side Refrigerant Side

Refrigerant R410A

Type TXV Inlet P+T, Outlet P+SH

TXV/EXV Inlet

Pressure 2.783 MPa Temperature 40.75 °C

Heat Exchange Inlet

Sat. T 0 K SH(+) 0 K

Pressure 0.000 Pa Temperature 0 K

Mass Flow 0 kg/s Quality 0

Heat Exchange Outlet

Pressure 1.083 MPa SH(+) / SC(-) 4.37 K

☒ Ref. in Single Flow

☐ Ref. in Parallel Flow with Same Out SH and Out P

☐ Ref. in Parallel Flow with Same Out P

(b) Refrigerant side

Figure 3.4 Boundary Condition Settings

The refrigerant side includes:

1. Refrigerant selection, which can be selected in the "Refrigerant" drop-down menu.
2. Types of refrigerant input:
 - a) Inlet P+T+M: For condenser, the MCHE inlet pressure, temperature and mass flow rate are known;
 - b) Inlet P+X+M: For evaporator, the MCHE inlet pressure, quality and mass flow rate are known;
 - c) Inlet P+X, Outlet SC: For condenser, the MCHE inlet pressure, quality and Outlet SC are known;
 - d) Inlet P+X, Outlet SH: For evaporator, the MCHE inlet pressure, quality and Outlet

SH are known

- e) TXV Inlet P+T, Outlet P+SH: For evaporator, the pressure and temperature before TXV, and the MCHE outlet pressure and outlet SH are known;
 - f) Inlet Tsat+SH+M: For condenser, the MCHE inlet saturation temperature, SH and mass flow rate are known;
 - g) Inlet Tsat+SH, Outlet SC: For condenser, the MCHE inlet saturation temperature, SH and Outlet SC are known.
3. Ref. flow type: in single direction flow or in parallel flow.

The dialog box 'NonUniform Set' has a 'Specified Velocities' section. It contains a table with columns for Segment_1, Segment_2, Segment_3, Segment_4, and Segment_5. The rows represent Tube_1 through Tube_12. All cells in the table are currently set to 0. There are buttons for 'Load from Excel', 'Load from Text', and 'Unit: m/s'.

	Segment_1	Segment_2	Segment_3	Segment_4	Segment_5
Tube_1	0	0	0	0	
Tube_2	0	0	0	0	
Tube_3	0	0	0	0	
Tube_4	0	0	0	0	
Tube_5	0	0	0	0	
Tube_6	0	0	0	0	
Tube_7	0	0	0	0	
Tube_8	0	0	0	0	
Tube_9	0	0	0	0	
Tube_10	0	0	0	0	
Tube_11	0	0	0	0	
Tube_12	0	0	0	0	

Figure 3.5 Non-Uniform Velocity Field

3.5 Correlation Page

Figure 3.6 shows the correlation selection UI. The current embedded correlations are shown in Table 1. Among them, “Wet SanHua” on the air side is the correlation for evaporator developed based on our own products and is recommended with auto-correction.

The 'Correlation' tab in the 'Other Setup' dialog box shows the following settings:

- Air Side:**
 - Heat Transfer: LouverFin: Wet SanHua (1, Auto-Correction checked)
 - Pressure Drop: LouverFin: Wet SanHua (1)
- Refrigerant Side:**
 - Single-Phase:**
 - Heat Transfer: Gnielinski (1)
 - Pressure Drop: Churchill (1)
 - Two-Phase:**
 - Heat Transfer: EvaKandikar-Balasubramanian (1)
 - Pressure Drop: Fritzel (0.6)
- DeHumidification Model:**
 - Enable DeHum. Model: checked
 - DeHum. Model: Condensate Retention/Water Film Thickness (1)

Figure 3.6 Correlation Selection UI

Table 1 Embedded Correlations

Classification		Correlations
Air Side	HTC	LouverFin: Kim-Bullurd_HTC
		LouverFin: Chang-Wang_HTC

		LouverFin: Wet SanHua
		Constant Value
	Pressure Drop	LouverFin: Kim-Bullurd_DP
		LouverFin: Chang-Wang_DP
		LouverFin: Wet SanHua
Refrigerant Side	Single Phase HTC	Gnielinski
		PengPetersonLaminar
		laminarRec
		Constant Value
	Single Phase Pressure Drop	Churchill
		LaminarRec
	Two Phase HTC	Con: Dobson
		Con: Shah2016
		Con: Cavallini
		Con: SanHua
		Eva:Kandlikar-Balasubramanian
		Eva:Jung-Radermacher
		Eva:Shah2014CO2
		Eva:Yun2006R410A
		Kandlikar
		Shah1982
		Constant Value
	Two Phase Pressure Drop	Fridel
		ChenEtalR410A
		Evap:ParkHrnjak
		Evap:Jung-Radermacher

3.6 Other Setting

Figure 3.7 shows the remaining settings, including the selection of non-uniform geometry types, as described in section 2.6, and the input of material density and price, which are used to calculate the weight and cost of MCHE.

Solver	Geometry	Pass	Boundary	Correlation	Other Setup																
NonUniform Geometry Setting ☒ With Same Air Face Area (N Banks) ☐ Folded MCHE With Different Air Face Area (2 Banks) ☐ User Defined Location (2 Banks) along Tube Length m along Tube Height m																					
Materials Density <table border="1"> <tr> <td>Tube</td> <td>Al</td> <td>2700</td> <td>kg/m3</td> </tr> <tr> <td>Fin</td> <td>Al</td> <td>2700</td> <td>kg/m3</td> </tr> <tr> <td>Header</td> <td>Al</td> <td>2700</td> <td>kg/m3</td> </tr> <tr> <td>Connecting Tube</td> <td>Copper</td> <td>8940</td> <td>kg/m3</td> </tr> </table>						Tube	Al	2700	kg/m3	Fin	Al	2700	kg/m3	Header	Al	2700	kg/m3	Connecting Tube	Copper	8940	kg/m3
Tube	Al	2700	kg/m3																		
Fin	Al	2700	kg/m3																		
Header	Al	2700	kg/m3																		
Connecting Tube	Copper	8940	kg/m3																		
Cost <table border="1"> <tr> <td>Tube</td> <td>0</td> <td>RMB/kg</td> </tr> <tr> <td>Fin</td> <td>0</td> <td>RMB/kg</td> </tr> <tr> <td>Header</td> <td>0</td> <td>RMB/kg</td> </tr> <tr> <td>Connecting Tube</td> <td>0</td> <td>RMB/kg</td> </tr> </table>						Tube	0	RMB/kg	Fin	0	RMB/kg	Header	0	RMB/kg	Connecting Tube	0	RMB/kg				
Tube	0	RMB/kg																			
Fin	0	RMB/kg																			
Header	0	RMB/kg																			
Connecting Tube	0	RMB/kg																			

Figure 3.7 Other Settings

3.7 Run

After completing the above settings, user can view the total input data on the left side of the main UI. Click "Run" on the toolbar to start the simulation and the entire convergence process can be seen on the main UI. After the calculation is completed, a prompt appears, as shown in Figure 3.8, then click OK to close it.

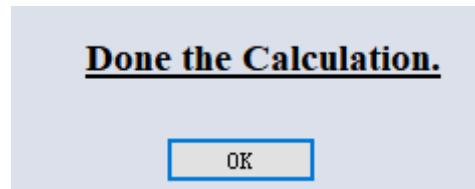


Figure 3.8 Simulation Completion Prompt

After completing the calculation, the user can click "Check Correlation" to check the applicability of the correlation, as shown in Figure 3.9.

Log

```

0 0.102358 1.083e+06 5.14021 1.06823e+06 0.77021 -14770.8
1 0.0989241 1.09695e+06 4.15731 1.08299e+06 -0.212688 -12.1113
2 0.0986236 1.09692e+06 4.36813 1.083e+06 -0.00187467 -3.26392
-----A Quick Look at Results-----
The Total Q is: 15964.4 W
The Ref. Side HTC is: 2182.62 W/m2K
The Air Side HTC is: 117.975 W/m2K
The Air DP is: 26.6269 Pa
The Ref. DP is: 13925.1 Pa
Check the applicability of Correlations...
Warning 1.1.1: The Ref.Two-Phase HTC Formula may not Suitable for: R410A.

```

[Check Correlations](#)

Start
Stop
Clear

Fig 3.9 Check Correlation